MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG4DSCECT200 PYTHON PROGRAMMING LAB

Duration: 2hours **Maximum Marks: 35**

Evaluation Criteria

- Viva - 10
- Lab report - 10
- Demonstration - 15

Task: (Select any one Exercise**)**

1 **Programs based on data types, Input & Output and Control Statements**

a) Find the factorial of a number N using a loop.

OR

b) Write a program to calculate the sum of the digits of a multi-digit integer

2 **Programs based on Arrays**

a) Count the total number of even and odd integers in an array.

OR

b) Merge two sorted arrays into a third array while keeping it sorted.

3 **Programs based on Strings**

a) Write a program to convert all characters of a string from lowercase to uppercase and vice versa.

OR

b) Find the frequency of a specific character (Eg: how many times 'e' appears in a sentence).

4 **Programs based on Functions**

a) Write a function calculate_circle(radius) that returns both the area and circumference as a tuple.

OR

b) Write a recursive function decimal_to_binary(n) to print the binary representation of a decimal number

5 **Programs based on Lists and Tuples**

a) Given a list of integers, create two separate lists: one containing all even numbers and another containing odd numbers.

OR

b) Create a program to pack student details (Name, Roll No, Grade) into a tuple and then unpack them into individual variables.

6 **Programs based on Dictionaries**

a) Write a program to generate and print a dictionary that contains numbers between 1 and nas keys and their squares as values.

OR

b)Write a program to count the number of occurrences of each character in a given string and store them in a dictionary.

7 **Programs to demonstrate basic plots using matplotlib**

a) Create a plot of $y= x^2$ for a range of values. Include a title, x-label ("Numbers"), and y-label ("Squares").

OR

b) Plot the sin(x) and cos(x) functions on the same graph. Use different colors

OR

c) Create a bar graph representing the marks of 5 students in a specific subject. Customize the bar colors and add labels on top of each bar.

8 **Programs based on Exception handling**

a) Write a program that takes a student's age as input. Use the raise keyword to trigger a Value Error if the age is negative or exceeds 100

OR

b) Define a user-defined exception named Insufficient Balance Error for a simple banking program that triggers when a withdrawal amount exceeds the balance.

9 **Programs based on Regular Expressions**

a) Check if a given string is a valid Indian mobile number (starts with 6-9 and has exactly 10 digits).

OR

b) Validate a PAN number (Format: 5 Uppercase letters, 4 digits, 1 Uppercase letter).

10 Make a login form with tkinter